

Yanshu Shingala

AI / ML & FULL-STACK DEVELOPER

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Portfolio

EDUCATION

JG University

Sep 2025 – Present

B.Tech in Data Science & Artificial Intelligence

Marwadi University

Sep 2022 – May 2025

Diploma in Computer Engineering — CGPA: 6.97

- Runners-Up, Code Storm inter-college coding competition

PROJECTS

LAIKA — Local AI Knowledge Assistant

2025

- Built a **fully offline RAG (Retrieval-Augmented Generation) system** that lets users query their own documents (PDF, DOCX, TXT, MD) with zero data leaving the machine.
- Implemented **hybrid search** combining ChromaDB semantic embeddings and SQLite FTS5 full-text search via Reciprocal Rank Fusion (RRF), improving retrieval relevance over single-method approaches.
- Architected a **FastAPI backend + Streamlit frontend** with multi-turn conversation memory, citation display, and full Docker Compose deployment support.
- Integrated local Ollama LLMs (qwen2.5:3b) for inference — no API keys or internet required.

Python

FastAPI

ChromaDB

SQLite FTS5

Streamlit

Ollama

Docker

RAG

Vaani — Offline Multilingual Voice Assistant

2025

- Engineered a **100% offline voice assistant** supporting Hindi, Gujarati, and English — no API keys, no cloud, all audio stays on-device.
- Built a **5-stage AI pipeline**: WebRTC noise cancellation → Faster-Whisper STT → Meta NLLB-200 translation → Ollama LLM → offline SAPI5 TTS.
- Implemented real-time silence detection and background-threaded TTS so the assistant feels responsive with no blocking waits.

Python

Faster-Whisper

Meta NLLB-200

Ollama

WebRTC VAD

Streamlit

NLP

House Price Predictor

2025 · Deployed on Render

- Trained a scikit-learn regression model on the Ames Housing dataset achieving $R^2 = 0.91$, $MAE = \$15,035$ on held-out test data.
- Built and deployed a **Flask REST API** with a /predict endpoint, full input validation, and Gunicorn production server via Procfile — live on Render.

Python

scikit-learn

Flask

Pandas

NumPy

Gunicorn

Render

Email Spam Detector

2024

- Built a full-stack ML web app using **TF-IDF vectorization + logistic regression** to classify emails as spam or legitimate with a live web interface.

[Python](#)[scikit-learn](#)[TF-IDF](#)[Vercel](#)

SKILLS

Languages	Python, SQL, JavaScript, C++, HTML/CSS
AI / ML	scikit-learn, Faster-Whisper, Meta NLLB-200, Ollama, RAG, TF-IDF, Prompt Engineering
Backend	Flask, FastAPI, REST APIs, Gunicorn
Frontend	React, Streamlit, Tailwind CSS
Databases	ChromaDB, SQLite, PostgreSQL
Tools	Git, Docker, Vercel, Render, Google Cloud

CERTIFICATIONS — GOOGLE CLOUD (ARCADE PROGRAM)

Create ML Models with BigQuery ML	Credly ↗
Prepare Data for ML APIs on Google Cloud	Credly ↗
Store, Process, and Manage Data on Google Cloud — Console	Credly ↗
Use Machine Learning APIs on Google Cloud	Credly ↗

ACHIEVEMENTS

Runners-Up — Code Storm, Marwadi University	2024
Inter-college competitive programming competition	

Microsoft Copilot for Productivity — Microsoft & LinkedIn Learning

Applied AI productivity tools across M365 (Word, Excel, PowerPoint, Outlook, Teams)

[Microsoft & linkedin](#)[↗](#)